A First Course in Group Theory

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Preface

The aim of this book is to provide a readable account of the examples and fundamental results of groups from a theoretical point of view and a geometrical point of view. The concept of a group is one of the most fundamental in modern mathematics. Groups are systems consisting of a set of elements and a binary operation that can be applied to two elements of the set, which together satisfy certain axioms. These require that the group is closed under the operation (the combination of any two elements produces another element of the group), that it obey the associative law, that it contains an identity element (which, combined with any other element, leaves the latter unchanged), and that each element have an inverse (which combines with an element to produce the identity element).

Since the book does not make any assumptions about the reader's background, it is suitable for newcomers to group theory and even those who never studied algebra. To explain many subjects we used figures and images to help the readers.

The book is organized into eleven chapters. To get to any depth in group theory requires set theory, combinatorics, number theory, matrix theory, and geometry. Thus, not only we have devoted Chaps. 1–2 to the introductory concepts of set theory, combinations, number theory, and symmetry, but we have also devoted some parts of Chap. 7 to matrix theory. Although the main subject of the book is groups but in many parts we will need some information about other algebraic structures, like rings, fields and vector spaces. In Chap. 3, we give the definitions of group and subgroup, examples and some elementary properties. Several excellent and important examples of groups like cyclic groups, permutation groups, group of arithmetical functions, matrix groups and linear groups are investigated in Chaps. 4–7. In Chap. 8, Lagrange's theorem as the most important theorem in finite groups is discussed. Chapters 9–11 address the normal subgroups, factor groups, derived subgroup, homomorphism, isomorphism and automorphism of groups. A consequence is Cayley's theorem in which every group could be realized as a permutation group.

Some chapters or sections are labelled as optional; this means that the readers can ignore them for the first study. Each section ends in a collection of exercises. The purpose of these exercises is to allow students to test their assimilation of the material,

to challenge their knowledge and ability. Moreover, at the end of each chapter we have two special sections: (1) Worked-Out Problems; (2) Supplementary Exercises.

In each worked-out problems section, I decided to try teach by examples, by writing out to solutions to problems. In choosing problems, three major criteria have been considered, to be challenging, interesting and educational. Moreover, there are many exercises at the end of each chapter as supplementary exercises. They are harder than before and serve to present the interesting concepts and theorems which are not discussed in the text.

The list of references at the end of the book is confined to works actually used in the text.

Yazd, Iran Bijan Davvaz

Contents

1 Preliminaries Notions 1
- 1.1 Sets and Equivalence Relations 1
- 1.2 Functions 8
- 1.3 Ordered Sets 11
- 1.4 Combinatorial Analysis 14
- 1.5 Divisibility and Prime Numbers 18
- 1.6 Worked-Out Problems 25
- 1.7 Supplementary Exercises 28

2 Symmetries of Shapes 31
- 2.1 Symmetry 31
- 2.2 Translations 34
- 2.3 Rotation Symmetries 36
- 2.4 Mirror Reflection Symmetries 38
- 2.5 Congruence Transformations 41
- 2.6 Worked-Out Problems 43
- 2.7 Supplementary Exercises 45

3 Groups 47
- 3.1 A Short History of Group Theory 47
- 3.2 Binary Operations 49
- 3.3 Semigroups and Monoids (Optional) 53
- 3.4 Groups and Examples 59
- 3.5 Turning Groups into Latin Squares (Optional) 72
- 3.6 Subgroups 76
- 3.7 Worked-Out Problems 81
- 3.8 Supplementary Exercises 85

4 Cyclic Groups 87
- 4.1 Group of Integers Modulo n 87
- 4.2 Cyclic Groups 91
- 4.3 Generating Sets 99

4.4 Worked-Out Problems . . . 101
4.5 Supplementary Exercises . . . 102

5 Permutation Groups . . . 105
5.1 Inverse Functions and Permutations . . . 105
5.2 Symmetric Groups . . . 108
5.3 Alternating Groups . . . 119
5.4 Worked-Out Problems . . . 124
5.5 Supplementary Exercises . . . 127

6 Group of Arithmetical Functions (Optional) . . . 131
6.1 Arithmetical Functions . . . 131
6.2 Dirichlet Product and Its Properties . . . 136
6.3 Multiplicative Functions . . . 139
6.4 Worked-Out Problems . . . 143
6.5 Supplementary Exercises . . . 147

7 Matrix Groups . . . 149
7.1 Introduction to Matrix Groups . . . 149
7.2 More About Vectors in $\mathbb{R}^n$ . . . 166
7.3 Rotation Groups . . . 170
7.4 Reflections in $\mathbb{R}^2$ and $\mathbb{R}^3$ . . . 175
7.5 Translation and Scaling Matrices . . . 180
7.6 Dihedral Groups . . . 183
7.7 Quaternion Group . . . 189
7.8 Worked-Out Problems . . . 191
7.9 Supplementary Exercises . . . 193

8 Cosets of Subgroups and Lagrange's Theorem . . . 195
8.1 Cosets and Their Properties . . . 195
8.2 Geometric Examples of Cosets . . . 198
8.3 Lagrange's Theorem . . . 202
8.4 Index of Subgroups . . . 207
8.5 A Counting Principle and Double Cosets . . . 209
8.6 Worked-Out Problems . . . 212
8.7 Supplementary Exercises . . . 214

9 Normal Subgroups and Factor Groups . . . 217
9.1 Normal Subgroups . . . 217
9.2 Factor Groups . . . 222
9.3 Cauchy's Theorem and Class Equation . . . 225
9.4 Worked-Out Problems . . . 229
9.5 Supplementary Exercises . . . 231

10 Some Special Subgroups 233
10.1 Commutators and Derived Subgroups 233
10.2 Derived Subgroups of Some Special Groups 237
10.3 Maximal Subgroups 240
10.4 Worked-Out Problems 243
10.5 Supplementary Exercises 245

11 Group Homomorphisms 247
11.1 Homomorphisms and Their Properties 247
11.2 Isomorphism Theorems 254
11.3 Cayley's Theorem 262
11.4 Automorphisms 265
11.5 Characteristic Subgroups 268
11.6 Another View of Linear Groups 270
11.7 Worked-Out Problems 273
11.8 Supplementary Exercises 279

References 283

Index 287

About the Author

Bijan Davvaz is Professor at the Department of Mathematics, Yazd University, Iran. Earlier, he served as the Head of the Department of Mathematics (1998–2002), Chairman of the Faculty of Science (2004–2006), and Vice-President for Research (2006–2008) at Yazd University, Iran. He earned his Ph.D. in Mathematics with a thesis on "Topics in Algebraic Hyperstructures" from Tarbiat Modarres University, Iran, and completed his M.Sc. in Mathematics from the University of Tehran, Iran. His areas of interest include algebra, algebraic hyperstructures, rough sets and fuzzy logic. On the editorial boards for 25 mathematical journals, Prof. Davvaz has authored 6 books and over 600 research papers, especially on algebra, fuzzy logic, algebraic hyperstructures and their applications.

Notation

$G, H, \ldots$	sets, groups, ...
$\in, a \in A$	membership
$\emptyset$	empty set
$\{x \mid p(x)\}$	set of all x that $p(x)$
$A \subseteq B$	A is a subset of B
$A \subset B$	A is a proper subset of B
$A_1 \cap A_2 \cap \cdots \cap A_n$ and $\bigcap_{i \in I} A_i$	intersection of sets
$A_1 \cup A_2 \cup \cdots \cup A_n$ and $\bigcup_{i \in I} A_i$	union of sets
$A - B$	difference set of A and B
$\mathbb{N}$	natural numbers
$\mathbb{Z}$	integers, additive group and ring of integers
$\mathbb{Q}$	field of rational numbers
$\mathbb{R}$	field of real numbers
$\mathbb{C}$	field of complex numbers
$\mathbb{F}$	field
$\mathbb{F}^*$	if $\mathbb{F}$ is a field, $\mathbb{F}^* = \mathbb{F} - \{0\}$
$R[x]$	ring of polynomials over a ring R
$\mathbb{Z}_n$	ring and group of integers modulo n
U_n	group of units modulo n
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$a \equiv b (\mathrm{mod}\, n)$	congruence modulo n
(a, b)	greatest common divisor of integers a and b
$f : A \to B$	a function of A to B
$f(a)$	image of a under f
$f(A)$	image of A under f
Imf	image of f
$f^{-1}(B)$	inverse image of B under f
id_X	the identity function $X \to X$
φ	Euler function
μ	Möbius function

f^{-1}	the inverse of the function f
$f \circ g$	composite function
$\binom{n}{k}$	binomial coefficient $n!/k!(n-k)!$
$\|A\|$	number of elements in a finite set A
$\|G\|$	order of the group G
$\circ(a)$	order of element a
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
a^{-1}	the inverse of a
$Z(G)$	center of G
$C_G(a)$	centralizer of a in G
$C_G(X)$	centralizer of the set X in G
$N_G(X)$	normalizer of the X in G
$\langle a \rangle$	cyclic group generated by a
$\langle X \rangle$	subgroup generated by a set X
AB	$\{ab \mid a \in A \text{ and } b \in B\}$
$A = (a_{ij})$	matrix whose entry in row i and column j is a_{ij}
$\det(A)$	determinant of matrix A
I_n	the identity $n \times n$ matrix
A^t	transpose of the matrix A
$tr(A)$	trace of the matrix A
$Mat_{m\times n}(\mathbb{F})$	the set of all $m \times n$ matrices over $\mathbb{F}$
aH	left coset
Ha	right coset
G/H	factor group
$[G : H]$	index of the subgroup H in the group G
$G \cong H$	G is isomorphic to H
C_n	cyclic group of order n
S_X	symmetric group on X
S_n	symmetric group of degree n
A_n	alternating group
D_n	dihedral group of order $2n$
Q_8	quaternion group
$GL_n(\mathbb{F})$	general linear group
$SL_n(\mathbb{F})$	special linear group
$O_n(\mathbb{F})$	orthogonal group
$SO_n(\mathbb{F})$	special orthogonal group
$UT_n(\mathbb{F})$	the set of upper triangular matrices such that all entries on the diagonal are non-zero
$LT_n(\mathbb{F})$	the set of lower triangular matrices such that all entries on the diagonal are non-zero
$\langle X, Y \rangle$	inner product of two vectors
$[x, y]$	$x^{-1}y^{-1}xy$

G'	derived subgroup of a group G
$Kerf$	Kernel of the homomorphism f
$End(G)$	the set of all endomorphism from G to itself
$Aut(G)$	automorphism group
$Inn(G)$	inner automorphism group